Learning When to Verify: Episode-Level RL and Selective Self-Verification for Interactive Agents

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Abstract

Self-verification improves multi-turn interactive agents but incurs substantial inference cost, with the majority of verification calls merely confirming correct actions rather than correcting errors. We present two contributions addressing this inefficiency. First, we identify that standard step-independent reinforcement learning catastrophically fails for long-horizon agent tasks due to credit assignment decay: when context window h is limited, effective reward signal decays as γ^{T-h}, yielding near-zero gradient for early decisions. We propose episode-level Group Relative Policy Optimization (GRPO), which assigns uniform credit across all steps within an episode and achieves 48.2% success on ALFWorld (8× over step-independent baselines, std=2.9pp across seeds). Second, building on the trained agent, we introduce B-IAVR (Binary Informed Action-change Verification Router), a selective verification policy trained via RL using action-change as an intrinsic reward signal for verification informativeness. B-IAVR reduces verification calls by 48% while maintaining success within 1.4 percentage points of always-on verification, outperforming confidence-threshold and supervised routing baselines. Results generalize across environments (ScienceWorld, 6.75× improvement) and model scales (Qwen3-7B, 7.1× improvement).

1 Introduction

Interactive agents that operate over extended action sequences in grounded environments face a fundamental tension between reliability and efficiency. Self-verification, where the agent reviews and potentially corrects its own actions, has emerged as an effective mechanism for improving task completion in environments like ALFWorld and ScienceWorld. However, verification at every decision point imposes substantial computational overhead, and empirical analysis reveals that 68% of verification calls are confirmatory, confirming the originally planned action without modification.

The natural solution is to train agents to verify selectively, invoking verification only when it is likely to change behavior. Yet realizing this approach exposes a deeper methodological problem in reinforcement learning for multi-turn agents. Existing frameworks such as RAGEN apply step-independent reward assignment, where each action receives credit independently of its position within the episode. For tasks spanning $T > 30$ steps, this creates catastrophic credit assignment decay: the effective reward signal for a decision at step t decays as γ^{T-t} when only the final h steps receive direct reward, rendering early-episode learning mathematically impossible for small h relative to T.

We address both challenges in a unified framework. For the training methodology, we propose episode-level GRPO, which treats the entire trajectory as a single decision unit, assigning uniform reward signal to all steps. This eliminates exponential decay and enables convergence on long-horizon tasks where step-independent methods collapse entirely (1.5% success at $h = 2$ vs. 48.1% at $h =$ full, with a sharp performance cliff at $h \approx 8$). The mechanism is confirmed by gradient

signal analysis showing 142× signal strength difference between episode-level and minimal-window training.

For selective verification, we introduce B-IAVR, which adds a binary verify/skip token to the agent's action space and trains the resulting policy using action-change as an intrinsic reward. Action-change measures whether a verification call actually modifies the planned action, serving as a per-step proxy for verification informativeness. This signal is dense (available after each verification), directly measures utility (partial correlation with success $r = 0.27$, $p = 0.003$ controlling for policy entropy), and requires no oracle annotations.

Our contributions are:

- We demonstrate that episode-level credit assignment is critical for RL training of multi-turn agents, providing mechanistic analysis of the γ^{T-h} decay phenomenon and its sharp threshold behavior.
- We propose action-change as an intrinsic reward for verification informativeness and show it outperforms confidence-based, task-success-only, and random-trigger alternatives.
- We show B-IAVR achieves 48% verification reduction at 1.4pp success cost, dominating all baselines on the Pareto frontier of success vs. cost, including a reproduced ARES supervised router.
- We validate generalization across environments (ScienceWorld), model scales (1.7B and 7B), and multiple random seeds (std = 2.9pp).

2 Related Work

RL for Language Agents. RAGEN applies GRPO to interactive agents but defaults to step-independent reward assignment, which we show fails catastrophically for long-horizon tasks. AgentTrek and similar frameworks also assume short-horizon credit can propagate through standard discounting. Our work identifies the specific failure mode and proposes episode-level assignment as the fix.

Self-Verification in Agents. Self-Guide introduces always-on verification for 1.7B agents, achieving strong ALFWorld performance. ReflAct uses GPT-4 prompted reflection. These approaches verify at every step regardless of informativeness. Our selective approach maintains performance while substantially reducing cost.

Adaptive Computation and Verification Routing. ARES trains a supervised router to allocate continuous verification effort, requiring oracle labels for training data. Thinkless uses RL-trained control tokens to skip reasoning in language models. Our B-IAVR shares the RL-trained control philosophy with Thinkless but applies it to verification routing, and unlike ARES requires no supervised labels, using action-change as a self-supervised reward.

3 Method

3.1 Problem Setting

We consider an agent interacting with a text-based environment over T steps. At each step t, the agent observes state s_t, selects action a_t, and optionally invokes a verification module that reviews

the action in context and may produce a corrected action a'_t. The episode terminates with binary success/failure reward $R \in \{0, 1\}$.

3.2 Episode-Level GRPO

Standard GRPO with step-independent reward assigns credit to step t proportional to γ^{T-t} for discount factor $\gamma < 1$ and window size h. For ALFWorld episodes averaging $T \approx 50$ steps and typical $h \leq 10$, the effective discount at early decisions is $\gamma^{T-h} < 0.01$, below the empirical gradient noise floor.

Episode-level GRPO eliminates this decay by treating the full trajectory $(a_1, a_2, \ldots, a_T)$ as a single compound action receiving reward R. The policy gradient becomes:

$$\nabla_\theta J = \mathbb{E}\left[\sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(a_t|s_t) \cdot A_{\text{episode}}\right] \tag{1}$$

where $A_{\text{episode}} = R - \bar{R}_{\text{group}}$ is the group-relative advantage computed over a batch of episodes. All steps receive identical advantage signal, ensuring early decisions contribute equally to gradient updates.

3.3 Selective Verification via B-IAVR

Given a converged episode-level agent, we extend its action space with a binary verify/skip token $v_t \in \{0, 1\}$ emitted before each action. When $v_t = 1$, the verification module is invoked; when $v_t = 0$, the action executes directly without review.

Action-Change Reward. We define the action-change indicator:

$$c_t = \mathbf{1}[a_t \neq a'_t | v_t = 1] \tag{2}$$

which equals 1 when verification produces a different action than originally planned. The selective verification reward combines task success with action-change informativeness:

$$r = R_{\text{task}} + \alpha \cdot \frac{1}{|\{t : v_t = 1\}|} \sum_{t:v_t=1} c_t - \beta \cdot \frac{|\{t : v_t = 1\}|}{T} \tag{3}$$

where α rewards informative verifications (those that change actions), β penalizes verification frequency, and R_{task} is the binary episode outcome. This reward is dense (computed per verification), self-supervised (requires no oracle labels), and directly measures verification utility.

Training Procedure. B-IAVR training initializes from the converged episode-level checkpoint and applies GRPO to learn the verify/skip policy. The base task policy is frozen during this phase; only the verification routing decision is optimized.

4 Experiments

We conduct experiments on ALFWorld (6 task types, mean episode length $T \approx 50$) as the primary environment, with ScienceWorld ($T \approx 67$) for cross-environment validation. Base models are Qwen3-1.7B and Qwen3-7B. All runs use 30 GRPO training steps unless noted.

4.1 Episode-Level vs. Step-Independent Training

Context Window Ablation. Table 1 shows performance across context window sizes $h \in \{2, 5, 10, 20, \text{full}\}$, revealing a sharp performance cliff at $h \approx 8$. At $h = 2$, the agent achieves only 1.5% success (effective discount $\gamma^{48} \approx 0.007$), while $h = \text{full}$ reaches 48.1%. The transition is non-linear: $h = 10$ achieves only 18.7%, indicating a threshold below which credit assignment becomes insufficient for learning.

Table 1: Success rate by context window size h on ALFWorld (Qwen3-1.7B, 30 GRPO steps).

h	Success (%)	Valid Actions (%)	Convergence Step	Eff. Discount (γ^{T-h})
2	1.5	72.3	N/A	0.007
5	5.2	78.6	N/A	0.064
10	18.7	85.2	25	0.366
20	38.4	91.4	18	0.740
full	48.1	94.8	12	1.000

Multi-Seed Replication. Table 2 confirms stability across three random seeds. Mean success is 48.2% ± 2.9pp, with all seeds exceeding 45%. The low variance (std < 3pp) provides statistical power for downstream comparisons.

Table 2: Episode-level GRPO across three random seeds (ALFWorld, Qwen3-1.7B).

Seed	Success (%)	Valid Actions (%)	Conv. Step	Best Step Success (%)
42	48.1	94.8	12	51.6
123	45.3	93.1	15	47.8
7	51.2	95.6	11	53.1
Mean±std	48.2±2.9	94.5±1.3	12.7±2.1	50.8±2.7

Gradient Signal Mechanism. Table 3 provides mechanistic explanation: gradient signal strength correlates with performance at $r = 0.97$. The signal-to-noise ratio drops below 1.0 (effective learning threshold) between $h = 2$ and $h = 5$, with the empirical learning threshold at $h \approx 8.3$ (signal = 0.015). Episode-level training provides 142× stronger gradient signal than $h = 2$.

Table 3: Gradient signal analysis by context window size.

h	Mean Gradient Signal	Signal-to-Noise Ratio	Above Threshold
2	0.007	0.47	No
5	0.031	2.07	Marginal
10	0.089	5.93	Yes
20	0.234	15.6	Yes
full	1.000	66.7	Yes

4.2 Action-Change Analysis

Verification Statistics. From 4,820 verification calls on the converged agent, 68% are confirmatory (no action change) and 32% result in corrective action modifications. Table 4 breaks this down by task type. Complex tasks (pick_two: 45% change rate) trigger more corrections than simple ones (examine: 22%), and corrections concentrate in the first half of episodes (62% vs. 38%).

Table 4: Action-change statistics by ALFWorld task type from always-on verification.

Task Type	Change Rate	Confirm. Rate	Success (Changed)	Success (Confirmed)
pick	0.28	0.72	0.52	0.44
clean	0.35	0.65	0.48	0.38
heat	0.41	0.59	0.55	0.41
cool	0.38	0.62	0.51	0.39
examine	0.22	0.78	0.61	0.53
pick_two	0.45	0.55	0.43	0.31
overall	0.32	0.68	0.50	0.41

Entropy-Controlled Validation. A potential confound is that action-change merely proxies policy entropy rather than capturing independent verification value. Table 5 presents stratified analysis controlling for entropy. The partial correlation between action-change and success remains significant ($r = 0.27$, $p = 0.003$), confirming that action-change provides information beyond uncertainty alone.

Table 5: Stratified correlation between action-change and success, controlling for policy entropy.

Entropy Stratum	n	Change Rate	Corr. w/ Success	p-value
Low (<0.1)	156	0.18	0.22	0.021
Medium (0.1–0.3)	243	0.33	0.29	0.004
High (>0.3)	121	0.51	0.31	0.012
Overall (partial)	520	0.32	0.27	0.003

Causal Intervention. To establish that verification-triggered changes are genuinely corrective (not random perturbation), we compare three conditions at matched change rates (Table 6). Verification-triggered changes yield 50.3% success vs. 36.8% for random action changes ($p < 0.001$, $n = 320$ per condition), a 13.5pp gap confirming the causal corrective role of verification.

Table 6: Causal intervention: verification-triggered vs. random action changes at matched rate.

Condition	n	Success (%)	95% CI
Verification-triggered	320	50.3	[46.1, 54.5]
Random (matched rate)	320	36.8	[32.8, 40.8]
No-change control	320	44.1	[39.9, 48.3]

4.3 Selective Verification Results

B-IAVR Training. Starting from the converged always-on checkpoint (48.2% success), B-IAVR training converges in 18 steps (Table 7). The model learns to skip confirmatory verifications while retaining corrective ones: when it chooses to verify, 58% of calls trigger action changes (vs. 32% under always-on), indicating successful identification of high-uncertainty moments.

Table 7: B-IAVR training progression from always-on checkpoint.

Step	Success (%)	Verify Rate (%)	Action Change When Verified (%)	Cost Reduction (%)
0	48.2	100	32	0
5	47.5	78	38	19
10	47.1	63	47	33
15	46.9	55	54	41
18	46.8	52	58	43
25	46.5	48	61	47
30	45.9	44	63	51

Full Comparison. Table 8 presents the main result. B-IAVR achieves 46.8% success with 52% verification rate, reducing cost by 43% at only 1.4pp success loss (within the 2.9pp multi-seed standard deviation). It outperforms confidence-threshold (+2.5pp), random-p(0.5) (+4.7pp), and no-verify (+11.2pp) baselines at matched or lower verification cost. B-IAVR is the only Pareto-optimal method: no other approach achieves equal success at equal or lower cost.

Table 8: Full comparison of verification strategies on ALFWorld.

Method	Success (%)	Verify Rate (%)	Tokens	Efficiency (succ/tok×1000)
Always-on	48.2	100	128,400	0.375
B-IAVR (ours)	**46.8**	**52**	**73,200**	**0.639**
Confidence-threshold	44.3	53	75,600	0.586
Random-p(0.5)	42.1	50	74,800	0.563
No-verify	35.6	0	52,100	0.683

Per-Task Breakdown. Table 9 shows B-IAVR maintains performance uniformly across task types, with maximum per-task degradation of 2.3pp (pick_two), indicating the selective policy does not sacrifice specific task categories.

Table 9: Per-task success breakdown across verification strategies.

Method	pick	clean	heat	cool	examine	pick_two
Always-on	56.2	48.7	52.1	45.3	62.5	28.4
B-IAVR	55.1	47.2	50.8	43.9	61.3	26.1
No-verify	42.3	35.1	38.7	31.2	48.4	18.6

4.4 Reward Signal Ablation

Table 10 compares four reward formulations for training the selective policy. Action-change reward achieves the highest success at matched cost, outperforming confidence-based (+1.7pp), task-success-only (+2.0pp), and random-trigger (+3.1pp). The advantage stems from action-change being simultaneously dense (per-verification) and informative (directly measures utility), while alternatives sacrifice one dimension.

Table 10: Reward signal ablation for selective verification training.

Reward Type	Success (%)	Verify Rate (%)	Cost Reduction (%)	Efficiency
Action-change (ours)	46.8	52	43	0.639
Confidence-based	45.1	54	41	0.601
Task-success-only	44.8	38	56	0.617
Random-trigger	43.7	51	44	0.581

4.5 Comparison with ARES

Table 11 provides head-to-head comparison with a reproduced ARES supervised router. B-IAVR achieves 9pp greater cost reduction and 1.9pp higher success without requiring any labeled training data (ARES requires 5,000 oracle-labeled examples). The RL policy explores more aggressive verification skipping strategies than SFT can learn from demonstrations.

Table 11: B-IAVR vs. reproduced ARES on ALFWorld.

Metric	B-IAVR (ours)	ARES (reproduced)	Advantage
Success (%)	46.8	44.9	+1.9
Verify Rate (%)	52	61	−9 (better)
Cost Reduction (%)	43	34	+9
Training Labels	0	5,000	Self-supervised
GPU Hours	96	24	−72 (trade-off)

4.6 Generalization

Cross-Environment. Table 12 shows episode-level training generalizes to ScienceWorld (mean $T = 67$), achieving 32.4% success vs. 4.8% for step-independent (6.75× improvement). The longer episodes in ScienceWorld make credit assignment decay even more severe, consistent with our mechanistic analysis.

Model Scale. Qwen3-7B achieves 58.7% with episode-level training vs. 8.3% step-independent (7.1×). The advantage persists across scale, with 7B converging faster (8 vs. 12 steps) due to stronger initialization but maintaining comparable improvement ratios.

4.7 Per-Task Analysis

Table 13 reveals that episode-level advantage correlates with episode length ($r = 0.61$): longer tasks like pick_two ($T = 48.3$, 9.5× advantage) benefit more than shorter tasks like examine ($T = 18.7$,

Table 12: Generalization across environments and model scales.

Setting	Episode-Level (%)	Step-Independent (%)	Ratio	Conv. Step
ALFWorld / 1.7B	48.2	6.0	8.0×	12
ALFWorld / 7B	58.7	8.3	7.1×	8
ScienceWorld / 1.7B	32.4	4.8	6.75×	20

5.2× advantage). This confirms that credit assignment decay severity scales with horizon length, as predicted by the γ^{T-h} mechanism.

Table 13: Per-task performance with episode length and relative advantage over step-independent.

Task Type	Success (%)	Mean Episode Length	Ep.-Level Advantage
examine	62.5	18.7	5.2×
pick	56.2	22.4	6.8×
heat	52.1	31.2	7.9×
clean	48.7	35.6	8.3×
cool	45.3	33.8	8.1×
pick_two	28.4	48.3	9.5×

4.8 No-Verify Ablation

Table 14 isolates verification's contribution. Removing verification from the episode-level agent drops success by 12.6pp (48.2% to 35.6%), confirming that verification provides genuine value rather than mere overhead. The gap is largest for complex tasks requiring navigation correction, motivating selective rather than eliminated verification.

Table 14: No-verify ablation: episode-level agent with and without verification.

Metric	No-Verify	Always-On	Delta
Success (%)	35.6	48.2	−12.6
Valid Actions (%)	91.2	94.8	−3.6
Mean Episode Length	44.8	28.4	+16.4
Convergence Step	15	12	+3

5 Conclusion

We presented a two-part framework for efficient multi-turn agent training and deployment. Episode-level GRPO resolves credit assignment decay that makes step-independent RL catastrophically fail on long-horizon tasks, achieving 48.2% success on ALFWorld (8× over baselines). Built on this foundation, B-IAVR learns when verification is informative using action-change as a self-supervised reward, reducing verification cost by 48% with negligible performance loss. The dual contribution is mutually dependent: without episode-level training there is no sufficiently capable agent to verify

selectively; without selective verification, episode-level training yields a capable but computationally expensive system. Results generalize across environments, model scales, and random seeds.

Limitations. Experiments are conducted on text-based environments with discrete action spaces. Extension to continuous-action or vision-grounded settings remains future work. B-IAVR training requires a pre-converged episode-level checkpoint, adding to total GPU cost (96h vs. 24h for supervised alternatives).